

# Seyed Ali Rashidi

## Computer Vision Engineer

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Visa Status: Student (Eligible for conversion to Work Permit / EU Blue Card)

### Profile

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Computer Vision Engineer with experience developing deep learning and image processing systems for real-world imaging applications. Skilled in CNN-based models, object detection, and image segmentation, with experience building end-to-end computer vision pipelines from data preparation to deployment. Currently serving as Payload Supervisor on the IGNIS CubeSat mission, focusing on thermal-infrared imaging performance, radiometric modeling, and Signal-to-Noise Ratio (SNR) analysis for spaceborne sensors. Experienced in Python, PyTorch, OpenCV, and containerized deployment using Docker and Google Cloud Run.

### Skills

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**Computer Vision** — Image classification, object detection, image segmentation, convolutional neural networks (CNNs), image preprocessing and augmentation, feature extraction, OpenCV-based image processing, image quality analysis

**Machine Learning & Deep Learning** — Supervised learning, PyTorch, TensorFlow / Keras, Model evaluation (mAP, IoU, precision/recall), Hyperparameter tuning, Performance optimization

**Imaging Systems & Sensors** — Thermal infrared imaging, Radiometry, Signal-to-Noise Ratio (SNR), Field of View (FoV), Microbolometric sensors, Edge-constrained imaging systems

**Deployment & MLOps** — Docker, REST inference APIs, Google Cloud Platform (GCP), Google Cloud Run, Model packaging, Containerized deployment, Model versioning, CI/CD

**Software Engineering** — Version control (Git), Software testing, debugging, and performance troubleshooting, Code review workflows, Linux (bash / csh)

**Programming & Scientific Computing** — Python, C++, NumPy, Pandas, SciPy, Matplotlib, Jupyter Notebook, Arduino, Zemax OpticStudio

**Soft Skills** — Analytical problem solving, Engineering judgment, Cross-disciplinary collaboration, Technical communication & documentation, Leadership & project coordination

### Professional Experience

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03/2025 – Present

#### University of Naples “Federico II” - IGNIS CubeSat Mission

##### Payload Supervisor – Computer Vision & Imaging

Payload Supervisor responsible for evaluating the imaging performance of a thermal-infrared Earth observation payload for a CubeSat mission.

- Developed physics-aware imaging performance models to predict Signal-to-Noise Ratio (SNR) and expected image quality for thermal-infrared Earth-observation imagery.
- Performed system-level simulations to evaluate Field of View (FoV), spatial resolution, ground footprint, and expected signal levels under realistic orbital conditions.
- Translated mission analysis data into imaging performance predictions, enabling quantitative assessment of observation feasibility.
- Applied time-series anomaly detection and trend analysis to mission simulation outputs, identifying non-optimal launch timing.
- Based on SNR trends and seasonal orbital effects, recommended postponing launch (~5 months) to improve early-mission imaging quality.
- Supported payload design trade-offs and engineering decision-making in collaboration with multidisciplinary spacecraft teams.

01/2016 – 12/2023

#### EZ-Tech

##### Data Recovery Engineer

Worked on low-level data recovery and diagnostics, focusing on signal integrity, failure analysis, and recovery strategies under hardware constraints. The role required structured problem solving, detailed analysis of corrupted or inaccessible data, and reliable execution in time- and quality-critical scenarios.

06/2020 – 12/2023

**Faraz Omran Tadbir**

**Part-Time IT support specialist**

Provided technical support for desktop systems and users in a Windows-based environment. Responsibilities included diagnosing hardware and software issues, resolving connectivity problems, and performing system setup and maintenance, with an emphasis on reliability and user support.

## Projects

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02/2026 – Present

**Brain Tumor Segmentation using YOLOv8 on Multimodal CT and MRI Imaging**

**Computer Vision / Deep Learning Project (Ongoing)**

Developing a deep learning-based medical imaging system for automated brain tumor segmentation using YOLOv8-seg on multimodal CT and MRI datasets.

- Designed and trained a YOLOv8 segmentation model using transfer learning with pretrained weights in a GPU-accelerated training environment.
- Implemented a computer vision pipeline for medical image preprocessing, training, and inference-based segmentation mask generation.
- Evaluated model performance using box mAP, mask mAP, precision, recall, loss curves, and confusion matrix analysis to assess convergence and generalization.
- Conducted systematic hyperparameter tuning and augmentation strategies to improve segmentation accuracy and cross-modality robustness.
- Implemented an inference pipeline for segmentation mask visualization, enabling qualitative assessment of tumor boundary detection.
- Ensured annotation consistency and dataset quality control across multimodal medical imaging datasets.

08/2024 – 11/2024

**Morphological Classification of Extragalactic Radio Sources Using CNN**

**Computer Vision / Deep Learning Project**

Developed a CNN-based computer vision system to classify extragalactic radio galaxies (FR0, FRI, FRII) using grayscale astronomical imaging datasets.

- Designed and implemented an end-to-end computer vision pipeline, including dataset preparation, preprocessing, class balancing, model training, and evaluation.
- Trained a Convolutional Neural Network (CNN) for automated morphological classification of radio galaxies.
- Achieved ~80–85% classification accuracy, with improved robustness after dataset balancing and augmentation strategies.
- Analyzed class-wise performance differences, identifying lower FRII accuracy due to dataset imbalance and morphological complexity.
- Packaged the trained model as a Dockerized REST inference service and deployed it on Google Cloud Run, enabling scalable HTTP-based prediction.
- Validated model portability and deployment stability through cloud-based inference testing.

07/2025 – 11/2025

**Thermal Infrared Heatmap & SNR Prediction**

Developed a physics-aware simulation model to predict imaging performance and Signal-to-Noise Ratio (SNR) for a thermal-infrared CubeSat payload using mission analysis inputs.

- Built a Python-based simulation framework to estimate thermal infrared signal levels and generate scenario-based SNR predictions.
- Modeled imaging performance as a function of orbital altitude, viewing geometry, and temperature contrast between target and background.
- Generated thermal heatmaps and performance distributions to evaluate imaging feasibility under varying mission conditions.
- Conducted sensitivity analysis to assess the impact of ground footprint, observation geometry, and temperature differences on image quality.
- Used simulation outputs to support payload design trade-offs and mission-level engineering decisions.
- Estimated expected imaging performance under realistic orbital and environmental constraints.

## Education

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09/2023 – present  
Naples, Italy

**University of Naples Federico II** 

**Master's Degree in Data Science**

Relevant coursework: Machine Learning, Computer Vision, Statistical Learning, Data Mining, Big Data, Advanced Statistical Modeling

Thesis : Signal-to-Noise Ratio characterization for thermal-infrared microbolometric imaging in CubeSat missions

09/2009 – 01/2012

**Islamic Azad University**

**BSc in Computer Software Engineering**

Core subjects: Programming, Data Structures, Algorithms, Systems Design, Engineering Mathematics

## Languages

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**English**  
B2

**Italian**  
A2

**Deutsch**  
A1

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